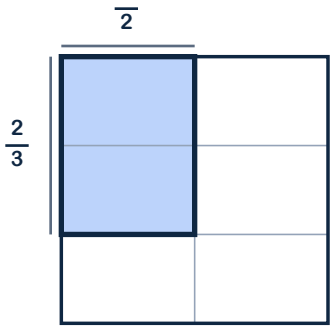
Multiplying Fractions

A fraction of a fraction: cut one way, then the other way.

1/2 × 2/3: half of two thirds



Cut the square into 3 rows: shade 2 of them (2/3).
Cut it into 2 columns: take half of the shaded part.
The square now has 6 pieces. 2 are in both.

$$\frac{1}{2} \times \frac{2}{3} = \frac{2}{6}$$

Without a picture: tops times tops, bottoms times bottoms

$$\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$$

2 × 4 = 8 pieces are in both.
3 × 5 = 15 pieces in the whole square.

Smaller or bigger?

$$\frac{3}{4} \times 8 = 6$$

Times a fraction less than 1: the answer is smaller than 8.

$$1\frac{1}{2} \times 8 = 12$$

Times more than 1: the answer is bigger than 8.

Your turn!

$$\frac{1}{3} \times \frac{3}{4} = \frac{3}{12}$$

For teachers

Goal: multiply fractions and see why it works (5.NF.4, 5.NF.5, 5.NF.6). Read × as "of": 1/2 × 2/3 is half of two thirds. Any equal fraction is right (2/6 or 1/3).

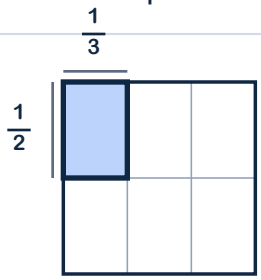
Multiplying Fractions: Practice

Read × as "of". Tops times tops, bottoms times bottoms.

Part 1 Read the picture

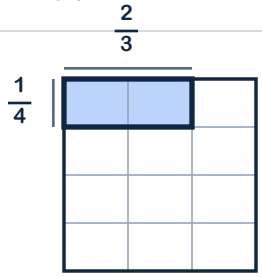
How many pieces are in both? How many pieces in all?

1)



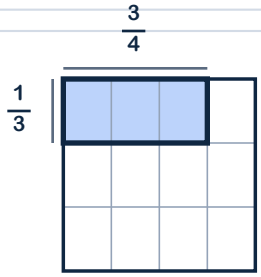
$\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$

2)



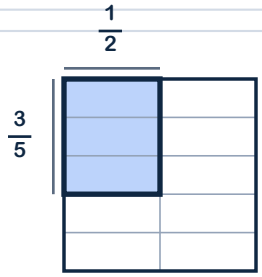
$\frac{2}{3} \times \frac{1}{4} = \frac{2}{12}$

3)



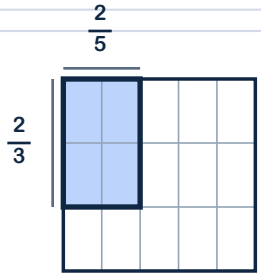
$\frac{3}{4} \times \frac{1}{3} = \frac{3}{12}$

4)



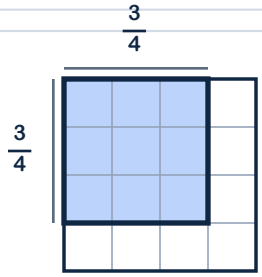
$\frac{1}{2} \times \frac{3}{5} = \frac{3}{10}$

5)



$\frac{2}{5} \times \frac{2}{3} = \frac{4}{15}$

6)

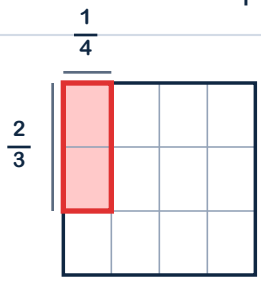


$\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$

Part 2 Shade and multiply

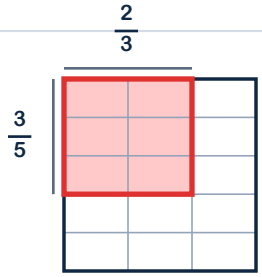
Shade the part that is in both, then write it.

7)



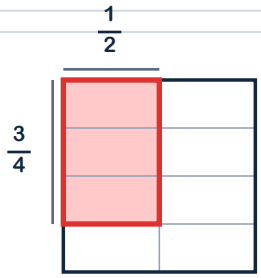
$\frac{1}{4} \times \frac{2}{3} = \frac{2}{12}$

8)



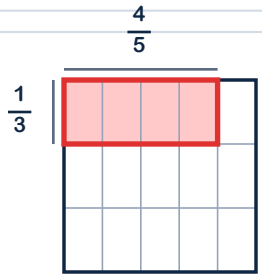
$\frac{2}{3} \times \frac{3}{5} = \frac{6}{15}$

9)



$\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$

10)



$\frac{4}{5} \times \frac{1}{3} = \frac{4}{15}$

Multiplying Fractions: Practice (continued)

Part 3 Multiply No help this time.

11) $\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$

12) $\frac{1}{4} \times \frac{3}{7} = \frac{3}{28}$

13) $\frac{3}{5} \times \frac{2}{9} = \frac{6}{45}$

14) $\frac{5}{6} \times \frac{1}{3} = \frac{5}{18}$

15) $\frac{3}{8} \times \frac{1}{2} = \frac{3}{16}$

16) $\frac{2}{7} \times \frac{3}{4} = \frac{6}{28}$

17) $\frac{4}{9} \times \frac{1}{5} = \frac{4}{45}$

18) $\frac{1}{6} \times \frac{5}{8} = \frac{5}{48}$

19) $\frac{3}{10} \times \frac{1}{4} = \frac{3}{40}$

20) $\frac{5}{7} \times \frac{2}{3} = \frac{10}{21}$

21) $\frac{1}{5} \times \frac{4}{5} = \frac{4}{25}$

22) $\frac{7}{8} \times \frac{1}{6} = \frac{7}{48}$

Multiplying Fractions: Practice (continued)

Part 3 Multiply (continued)

23) $\frac{2}{9} \times \frac{2}{5} = \frac{4}{45}$

24) $\frac{3}{4} \times \frac{5}{6} = \frac{15}{24}$

25) $\frac{1}{8} \times \frac{3}{10} = \frac{3}{80}$

26) $\frac{6}{7} \times \frac{1}{2} = \frac{6}{14}$

Part 4 A fraction of a number No help this time.

27) $\frac{2}{3} \times 9 = 6$

28) $\frac{3}{4} \times 12 = 9$

29) $\frac{4}{5} \times 15 = 12$

30) $\frac{5}{6} \times 18 = 15$

31) $\frac{3}{8} \times 56 = 21$

32) $\frac{4}{7} \times 14 = 8$

Multiplying Fractions: Practice (continued)

Part 4 A fraction of a number (continued)

33) $\frac{1}{6} \times 30 = 5$

34) $\frac{7}{10} \times 20 = 14$

35) $\frac{3}{5} \times 30 = 18$

36) $\frac{5}{8} \times 24 = 15$

Part 5 Mixed numbers times a number No help this time. Wholes first, then the piece.

37) $1\frac{1}{2} \times 4 = 6$

38) $2\frac{1}{3} \times 6 = 14$

39) $1\frac{3}{4} \times 4 = 7$

40) $3\frac{1}{2} \times 6 = 21$

41) $2\frac{2}{5} \times 5 = 12$

42) $2\frac{3}{4} \times 4 = 11$

43) $1\frac{1}{6} \times 6 = 7$

44) $4\frac{1}{3} \times 9 = 39$

Multiplying Fractions: Practice (continued)

Part 6 Smaller or bigger? No help this time. Write < or >. Don't work it out: think.

45) $\frac{3}{4} \times 8 \boxed{<} 8$

46) $1\frac{1}{2} \times 10 \boxed{>} 10$

47) $\frac{2}{3} \times 9 \boxed{<} 9$

48) $2\frac{1}{4} \times 20 \boxed{>} 20$

49) $\frac{5}{6} \times 12 \boxed{<} 12$

50) $1\frac{3}{5} \times 7 \boxed{>} 7$

51) $\frac{7}{8} \times 16 \boxed{<} 16$

52) $1\frac{1}{3} \times 15 \boxed{>} 15$

53) $\frac{4}{9} \times 18 \boxed{<} 18$

54) $3\frac{2}{7} \times 6 \boxed{>} 6$

Multiplying Fractions: Practice (continued)

Part 7 Multiply

55) $\frac{2}{3} \times \frac{2}{5} = \frac{4}{15}$

56) $\frac{3}{6} \times \frac{1}{3} = \frac{3}{18}$

57) $\frac{1}{4} \times \frac{8}{12} = \frac{8}{48}$

58) $\frac{7}{10} \times \frac{1}{2} = \frac{7}{20}$

59) $\frac{1}{8} \times \frac{8}{10} = \frac{8}{80}$

60) $\frac{8}{10} \times \frac{4}{6} = \frac{32}{60}$

61) $\frac{4}{10} \times \frac{7}{10} = \frac{28}{100}$

62) $\frac{5}{8} \times \frac{2}{5} = \frac{10}{40}$

63) $\frac{6}{10} \times \frac{1}{12} = \frac{6}{120}$

64) $\frac{2}{3} \times \frac{9}{12} = \frac{18}{36}$

65) $\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$

66) $\frac{11}{12} \times \frac{4}{6} = \frac{44}{72}$

Multiplying Fractions: Practice (continued)

Part 7 Multiply (continued)

67) $\frac{1}{10} \times \frac{2}{4} = \frac{2}{40}$

68) $\frac{2}{4} \times \frac{3}{4} = \frac{6}{16}$

69) $\frac{5}{12} \times \frac{5}{8} = \frac{25}{96}$

70) $\frac{2}{12} \times \frac{2}{3} = \frac{4}{36}$

71) $\frac{1}{4} \times \frac{1}{12} = \frac{1}{48}$

72) $\frac{3}{12} \times \frac{1}{2} = \frac{3}{24}$

73) $\frac{1}{2} \times \frac{5}{6} = \frac{5}{12}$

74) $\frac{4}{6} \times \frac{1}{3} = \frac{4}{18}$

75) $\frac{11}{12} \times \frac{4}{5} = \frac{44}{60}$

76) $\frac{1}{2} \times \frac{5}{10} = \frac{5}{20}$

77) $\frac{1}{2} \times \frac{7}{8} = \frac{7}{16}$

78) $\frac{1}{10} \times \frac{4}{6} = \frac{4}{60}$

Multiplying Fractions: Practice (continued)

Part 8 A fraction of a number

79) $\frac{2}{6} \times 18 = 6$

80) $\frac{1}{2} \times 12 = 6$

81) $\frac{2}{4} \times 8 = 4$

82) $\frac{1}{3} \times 6 = 2$

83) $\frac{2}{3} \times 6 = 4$

84) $\frac{1}{3} \times 15 = 5$

85) $\frac{3}{6} \times 24 = 12$

86) $\frac{1}{2} \times 8 = 4$

87) $\frac{4}{5} \times 20 = 16$

88) $\frac{1}{2} \times 10 = 5$

89) $\frac{2}{3} \times 15 = 10$

90) $\frac{1}{5} \times 10 = 2$

Multiplying Fractions: Practice (continued)

Part 8 A fraction of a number (continued)

91) $\frac{2}{6} \times 24 = 8$

92) $\frac{1}{4} \times 8 = 2$

93) $\frac{7}{8} \times 40 = 35$

94) $\frac{1}{3} \times 12 = 4$

95) $\frac{2}{3} \times 18 = 12$

96) $\frac{1}{6} \times 18 = 3$

Part 9 Remember? Problems from earlier packets.

97) $\frac{1}{3} + \frac{1}{2} = \frac{5}{6}$

98) $\frac{4}{5} - \frac{1}{4} = \frac{11}{20}$

99) $6 \times 0.7 = 4.2$

100) $0.4 \times 0.6 = 0.24$

101) $3 \times \frac{2}{7} = \frac{6}{7}$

102) $752 \div 16 = 47$

Multiplying Fractions: Practice (continued)

Part 10 Word problems × means "of".

103) A recipe needs 3/4 cup of milk. Mia makes half of the recipe. How much milk does she need?

Number sentence 1/2 × 3/4 = 3/8

Answer 3/8 cup

Draw or show your work

104) A garden is 2/3 of the yard. Tomatoes grow on 1/2 of the garden. What part of the yard is that?

Number sentence 1/2 × 2/3 = 2/6

Answer 2/6 of the yard

Draw or show your work

105) A class has 24 kids. 3/8 of them walk to school. How many kids walk?

Number sentence 3/8 × 24 = 9

Answer 9 kids

Draw or show your work

106) A ribbon is 1 1/2 m long. Leo needs 6 ribbons. How much ribbon is that?

Number sentence 1 1/2 × 6 = 9

Answer 9 m

Draw or show your work

107) Challenge: Ava had 4/5 of a pizza. She ate 1/4 of that. How much of the whole pizza did she eat?

Number sentence 1/4 × 4/5 = 4/20

Answer 4/20 of the pizza

Draw or show your work

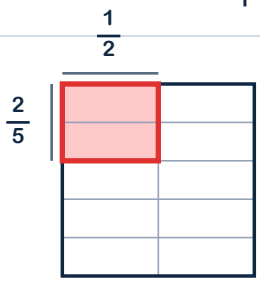
Check-In: Multiplying Fractions

Show what you know. No help this time.

Mastery: 9 of 11 (80%)

Part 1 Shade and multiply

1)



$\frac{1}{2} \times \frac{2}{5} = \frac{2}{10}$

Part 2 Multiply

2) $\frac{2}{5} \times \frac{3}{7} = \frac{6}{35}$

3) $\frac{3}{4} \times \frac{2}{5} = \frac{6}{20}$

4) $\frac{1}{2} \times \frac{7}{9} = \frac{7}{18}$

5) $\frac{4}{7} \times \frac{2}{3} = \frac{8}{21}$

Part 3 A fraction of a number

6) $\frac{3}{4} \times 20 = 15$

7) $\frac{5}{6} \times 24 = 20$

8) $\frac{3}{5} \times 10 = 6$

Part 4 Smaller or bigger? Write < or >.

9) $\frac{2}{5} \times 10 \; < \; 10$

10) $1\frac{1}{4} \times 8 \; > \; 8$

11) $\frac{5}{7} \times 14 \; < \; 14$